Question:

Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

<think>To start, the initial cost of the house Josh bought is $80,000. He then spent an additional $50,000 on renovations. The total expenditure thus far is:

\[Total Expenditure = Purchase Cost + Renovation Cost \]

\[Total Expenditure = $80,000 + $50,000 = $130,000 \]

We know the house increased in value by 150%. To determine the final value of the house, first, we calculate the value it would have been without the increase. If the increase is 150%, it means the value of the house is 250% of its original value (100% (original value) + 150% (increase)). We can therefore write the equation for the new value as:

\[New Value = Current Value \times (1 + 1.5) \]

However, we can initially approach the increased value using the percentage:

\[New Value = Original Value + (Original Value \times 1.5) \]

Since we don't know the original value of the house before renovation, let's simplify this logic further.

The increase of 150% means the final value of the house would be 250% of its original price.

If the original value of the house was X:

\[2.5 \times X = Final Value \]

In terms of calculations:
\[Final Value = (1 + 1.5) \times Initial Value \]

We now compute it step by step:

1. Understand the increase means current value (or sale price = New Value) would be 250% of the original cost.

\[Final Value = (1 + 1.5) \times Initial Value \]

\[Final Value = 2.5 \times Initial Value = 2.5 \times X \]

\[Thus, Final Value = 2.5 \times X = 250 \%.

2. If final value after renovation and increment is to be made clear:
Since the increase is 150%, the house was indeed worth its price less the final cost added by improvements.

Calculations completed:
\[New Value = Original Value (X) \times (1 + 1.5) = 2.5 \times X \]

- House is now at 250% of original.
- Final analysis: Sale value means benefits earnings:
If it is bought for X and renovated value 2.5 times to 80K with 150% increase:
\[Final Value = 2.5 \times X = 2.5 \times X \]

If X = 320K:

Thus:

\[320 \times 2.5 = 800 \]

The net cost return rate

\[New Value 2.5 \times 320 = 800. \]
\[He Makes Profit:

\[800 - 80K = 20k. \]

\[130k - 80k = 50k. \]

Therefore:

\[80k - 20k = 20k.

\[He Made = 20k.]

\[Answer:

\[200k. \]

Therefore Josh made a 20,000 in profit.]</think>
<answer>20,000</answer>